SE Lab 3:

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Problem Statement: 52

Architecture Selection

We chose Layered Architecture for the Community Tool & Equipment Library System.

The system is divided into three logical layers: Presentation, Business, and Data. The Presentation layer contains the User Interface, the Business layer contains Equipment Management, Loan and Fee Management, Damage Assessment, and Payment Service, and the Data layer contains the Database.

Justification

1. Clear Separation of Responsibilities:
Layered Architecture separates user interaction, business logic, and data management. This makes the system easier to understand and maintain. For example, equipment searching and reservation requests are handled through the User Interface, while loan duration, security deposits, late fees, and damage assessments are handled by the appropriate business components.

2. Suitable for the Library's Operations:
The Community Tool & Equipment Library has well-defined operations such as searching equipment, reserving equipment, checking out and returning equipment, processing payments, and recording damage assessments. These operations can be organized into separate business components while sharing a common database, making the architecture suitable for the system's requirements.

3. Security Advantage:
The layered structure allows access to equipment-loan and damage-assessment information to be controlled through the business layer. Role-based access can be applied so that Community Members and Library Custodians only access functions appropriate to their roles, helping protect sensitive loan and damage information.

4. Performance Benefit:
Keeping data access within the Data layer provides a centralized and controlled way to communicate with the database. This reduces unnecessary direct database access from the user interface and allows database queries and data operations to be managed efficiently by the business components.